IMAT 모의고사 4회

60 questions · 100 minutes · +1.5 / −0.4 / 0

Section 1 — Reading Skills & General Knowledge (Q1–4)

Passage for Q1–2

> Urban heat islands form because asphalt, concrete and dark roofs absorb sunlight during the day and release heat slowly at night, while cities have fewer trees to provide shade and evaporative cooling. During heatwaves, night-time temperatures in dense neighbourhoods can remain several degrees higher than in nearby countryside, and hospital admissions among older residents rise sharply. Planting street trees and using reflective roofing are among the cheapest interventions, but their benefits are unevenly distributed: wealthier districts tend to have more green space already.

1. Which of the following is given in the passage as a reason why cities stay warmer at night?

A. Older residents use more heating.
B. Building materials release absorbed heat slowly.
C. Hospitals generate heat.
D. Cities receive more sunlight than the countryside.
E. Reflective roofs trap heat.

2. What concern does the author raise in the final sentence?

A. The benefit of greening may not reach the neighbourhoods that need it most.
B. Reflective roofs are too expensive.
C. Trees increase night-time temperatures.
D. Wealthy districts suffer most from heatwaves.
E. Heat islands do not affect health.

3. Which organ was first successfully transplanted between humans (identical twins) in 1954?

A. Heart
B. Cornea
C. Kidney
D. Lung
E. Liver

4. In which year did the Berlin Wall fall?

A. 1975
B. 1985
C. 1961
D. 1991
E. 1989

Section 2 — Logical Reasoning & Problem Solving (Q5–9)

5. "No reptiles are mammals. All snakes are reptiles." Which conclusion is valid?

A. No mammals are reptiles and some are snakes.
B. All reptiles are snakes.
C. Some snakes are mammals.
D. Some mammals are snakes.
E. No snakes are mammals.

6. A nurse can prepare 12 infusion bags per hour; a second nurse can prepare 8 per hour. Working together, how long do they take to prepare 50 bags?

A. 2 h
B. 4 h 10 min
C. 4 h
D. 3 h
E. 2 h 30 min

7. A code replaces each letter with the letter two places later in the alphabet (A→C, B→D, ..., Y→A, Z→B). How is the word "ZEBRA" written?

A. XCZPY
B. BGDTA
C. AFCSB
D. BGDTC
E. BGDSC

8. The average of five numbers is 12. When a sixth number is added, the average becomes 13. What is the sixth number?

A. 13
B. 18
C. 20
D. 15
E. 25

9. A clinical trial has 240 participants divided between drug and placebo in the ratio 5 : 3. How many receive the placebo?

A. 80
B. 90
C. 48
D. 120
E. 150

Section 3 — Biology (Q10–32)

10. Which organelle contains hydrolytic enzymes that digest worn-out organelles?

A. Lysosome
B. Vacuole of animal cells
C. Centriole
D. Nucleolus
E. Ribosome

11. Which enzyme synthesises an RNA primer during DNA replication?
A. Reverse transcriptase
B. Primase
C. Ligase
D. Helicase
E. DNA polymerase I

12. Which checkpoint protein is often called "the guardian of the genome" and can halt the cell cycle when DNA is damaged?
A. Tubulin
B. Insulin receptor
C. p53
D. Cyclin B
E. Actin

13. In the absence of oxygen, human muscle cells convert pyruvate to:
A. ethanol and CO_2.
B. acetyl-CoA.
C. citrate.
D. lactate.
E. glucose.

14. In a pedigree, two unaffected parents have an affected daughter. What is the most likely mode of inheritance?
A. X-linked dominant
B. X-linked recessive
C. Y-linked
D. Autosomal dominant
E. Autosomal recessive

15. Which pigment absorbs light energy for photosynthesis in plants?
A. Haemoglobin
B. Chlorophyll
C. Melanin
D. Myoglobin
E. Carotene only

16. Gas exchange in the lungs takes place in the:
A. alveoli.
B. bronchioles.
C. trachea.
D. bronchi.
E. pleura.

17. Which of the following is a monosaccharide?
A. Amylose
B. Fructose
C. Maltose
D. Lactose
E. Sucrose

18. At rest, the membrane potential of a typical neuron is approximately:
A. +30 mV
B. −150 mV
C. +70 mV
D. −70 mV
E. 0 mV

19. The pacemaker of the heart is the:
A. aortic valve.
B. bundle of His.
C. atrioventricular node.
D. sinoatrial node.
E. Purkinje fibres.

20. In a cross Aa × aa, what proportion of offspring will be homozygous?
A. 1/4
B. 3/4
C. 1/2
D. 1
E. 0

21. Which gland secretes adrenaline?
A. Adrenal medulla
B. Pancreas
C. Thyroid
D. Pituitary
E. Adrenal cortex

22. Nucleotides are made of:
A. a fatty acid and glycerol.
B. an amino acid, a sugar and a phosphate.
C. a nitrogenous base, a pentose sugar and a phosphate group.
D. glucose and a phosphate group.
E. a nitrogenous base and an amino acid.

23. Which immunoglobulin class can cross the placenta?
A. IgA
B. IgE
C. IgD
D. IgG
E. IgM

24. Antidiuretic hormone (ADH) acts by:
A. increasing glomerular filtration rate.
B. stimulating renin release.
C. reducing water reabsorption in the collecting duct.
D. stimulating sodium excretion.
E. increasing water permeability of the collecting duct.

25. A cell has 24 chromosomes at the end of telophase II of meiosis. How many chromosomes did the parent cell have before meiosis began?

A. 48
B. 12
C. 72
D. 24
E. 96

26. Which of the following stabilises the α-helix of a protein's secondary structure?

A. Phosphodiester bonds
B. Disulfide bridges between distant regions
C. Hydrogen bonds between backbone C=O and N–H groups
D. Glycosidic bonds
E. Ionic bonds between side chains only

27. Sickle cell anaemia is caused by:

A. a single base substitution in the β-globin gene.
B. a frameshift mutation in insulin.
C. a deletion of a whole chromosome.
D. trisomy.
E. an X-linked dominant gene.

28. Which statement about osmosis is correct?

A. Water moves across a partially permeable membrane from a region of lower solute concentration to higher solute concentration.
B. Osmosis is the movement of solutes.
C. Water moves from a region of higher solute concentration to lower solute concentration.
D. Osmosis only occurs in plants.
E. Osmosis requires ATP.

29. Which method of horizontal gene transfer in bacteria requires direct cell-to-cell contact via a pilus?

A. Binary fission
B. Conjugation
C. Transduction
D. Transformation
E. Sporulation

30. Which enzyme, secreted in saliva, begins starch digestion?

A. Lactase
B. Trypsin
C. Amylase
D. Lipase
E. Pepsin

31. A polypeptide of 150 amino acids is coded for by a minimum of how many nucleotides of mRNA (including the stop codon)?

A. 300
B. 450
C. 600
D. 150
E. 453

32. A species that has evolved into several species, each adapted to a different niche, is an example of:

A. convergent evolution.
B. genetic drift.
C. co-dominance.
D. adaptive radiation.
E. extinction.

Section 4 — Chemistry (Q33–47)

33. What mass of NaCl (M = 58.5 g/mol) contains 2 mol of chloride ions?

A. 117 g
B. 234 g
C. 29.25 g
D. 87.75 g
E. 58.5 g

34. What is the oxidation number of sulfur in H_2SO_3?

A. +6
B. −2
C. +8
D. +4
E. +2

35. Mixing equal volumes of 0.2 mol/L HCl and 0.1 mol/L NaOH gives a solution with $[H^+]$ of:

A. 0.3 mol/L
B. 0.15 mol/L
C. 0.05 mol/L
D. 0.01 mol/L
E. 0.1 mol/L

36. Electronegativity generally:

A. is the same for all halogens.
B. increases across a period from left to right.
C. is highest for alkali metals.
D. decreases across a period from left to right.
E. increases down a group.

37. How many σ and π bonds are in a molecule of ethene, C_2H_4?

A. 6 σ, 0 π
B. 4 σ, 2 π
C. 5 σ, 1 π
D. 5 σ, 2 π
E. 4 σ, 1 π

38. In the reaction $2Al + 3Cl_2 \rightarrow 2AlCl_3$, how many moles of Cl_2 are needed to react with 6 mol of Al?

A. 12
B. 6
C. 9
D. 4
E. 3

39. Which reaction produces an ester?

A. Alcohol + oxygen
B. Alkane + chlorine
C. Alcohol + carboxylic acid (acid catalyst)
D. Aldehyde + hydrogen
E. Alkene + water

40. A solution contains 4.9 g of H_2SO_4 (M = 98 g/mol) in 250 mL. What is its molar concentration?

A. 0.1 mol/L
B. 0.4 mol/L
C. 0.2 mol/L
D. 0.05 mol/L
E. 2 mol/L

41. Which compound has the highest boiling point?

A. CH_4
B. All have the same boiling point
C. C_3H_8
D. C_4H_{10} (n-butane)
E. C_2H_6

42. A buffer is made from NH_3 and NH_4Cl. If a small amount of HCl is added, which species reacts with it?

A. NH_3
B. NH_4^+
C. H_2O
D. Cl^-
E. OH^- produced by NH_4^+

43. For a reaction at equilibrium, $K_c = 10^{-5}$. This indicates that:

A. the reaction is exothermic.
B. the reaction cannot occur.
C. the products are strongly favoured.
D. the reactants are favoured at equilibrium.
E. the reaction is very fast.

44. Which of these compounds is an amine?

A. CH_3CN
B. CH_3NO_2
C. CH_3COOH
D. CH_3NH_2
E. CH_3CONH_2

45. An element has electron configuration $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^5$. To which group does it belong?

A. Group 1
B. Group 2
C. Group 17
D. Group 15
E. Group 16

46. In the reaction $2H_2O_2 \rightarrow 2H_2O + O_2$, the oxygen in H_2O_2 is:

A. both oxidised and reduced (disproportionation).
B. neither oxidised nor reduced.
C. only reduced.
D. converted to ozone.
E. only oxidised.

47. Benzene is best described as:

A. an ionic compound.
B. a saturated hydrocarbon.
C. a cycloalkane with three isolated double bonds.
D. a linear alkyne.
E. an aromatic ring with delocalised π electrons.

Section 5 — Physics & Mathematics (Q48–60)

48. A cyclist travels 120 m in 8 s at constant speed, then 80 m in 12 s. What is the average speed for the whole trip?

A. 12 m/s
B. 10 m/s
C. 20 m/s
D. 15 m/s
E. 8 m/s

49. A spring with spring constant 200 N/m is stretched by 0.1 m. How much elastic potential energy is stored?

A. 20 J
B. 10 J
C. 0.5 J
D. 2 J
E. 1 J

50. A 4 Ω resistor and a 12 Ω resistor in parallel are connected in series with a 2 Ω resistor. What is the total resistance?

A. 14 Ω
B. 5 Ω
C. 18 Ω
D. 3 Ω
E. 8 Ω

51. If the frequency of a wave is doubled while its speed remains constant, its wavelength:

A. halves.
B. doubles.
C. becomes zero.
D. stays the same.
E. quadruples.

52. A hydraulic press has a small piston of area 0.01 m² and a large piston of area 0.5 m². A force of 100 N is applied to the small piston. What force is exerted by the large piston?

A. 2 N
B. 50 000 N
C. 500 N
D. 50 N
E. 5000 N

53. An object moves in a circle of radius 2 m at a constant speed of 4 m/s. What is its centripetal acceleration?

A. 2 m/s²
B. 32 m/s²
C. 4 m/s²
D. 16 m/s²
E. 8 m/s²

54. A lamp rated 60 W is used for 5 hours. How much energy does it use?

A. 3 kWh
B. 300 kWh
C. 12 kWh
D. 0.3 kWh
E. 0.012 kWh

55. Solve: 2^(x+1) = 32.

A. 16
B. 5
C. 6
D. 3
E. 4

56. If sin θ = 3/5 and θ is acute, what is tan θ?

A. 4/5
B. 3/4
C. 3/5
D. 4/3
E. 5/3

57. How many different ways can 4 different books be arranged on a shelf?

A. 4
B. 16
C. 8
D. 12
E. 24

58. Which of the following is the equation of a line parallel to $y = 3x + 2$ passing through $(0, -1)$?

A. $y = -3x - 1$
B. $y = -(1/3)x + 2$
C. $y = (1/3)x - 1$
D. $y = 3x - 1$
E. $y = 3x + 1$

59. Solve the inequality $3 - 2x > 7$.

A. $x < -2$
B. $x < 5$
C. $x > -2$
D. $x < 2$
E. $x > 2$

60. A radioactive sample has a half-life of 6 hours. What fraction remains after 24 hours?

A. 1/4
B. 1/16
C. 1/24
D. 1/8
E. 1/32

정답 및 해설

번호	정답	해설
1	B	아스팔트·콘크리트가 흡수한 열을 천천히 방출.
2	A	녹화 효과가 이미 녹지 많은 부유층 지역에 쏠림 → 형평성 우려.
3	C	1954년 일란성 쌍둥이 간 신장 이식(머리, 보스턴).
4	E	1989년.
5	E	뱀 ⊂ 파충류, 파충류 ∩ 포유류 = ∅ → 뱀은 포유류가 아님.
6	E	20개/h → 50/20 = 2.5 h.
7	D	Z→B, E→G, B→D, R→T, A→C → BGDTC.
8	B	13×6 − 12×5 = 78 − 60 = 18.
9	B	240 × 3/8 = 90.
10	A	리소좀(자가포식).
11	B	프라이메이스.
12	C	p53.
13	D	젖산 발효.
14	E	정상 부모 → 이환 딸: 상염색체 열성(X연관 열성이면 아버지가 이환이어야 함).
15	B	엽록소.
16	A	폐포.
17	B	과당.
18	D	약 −70 mV.
19	D	동방결절(SA node).
20	C	Aa × aa → Aa 1/2(이형), aa 1/2(동형) → 동형접합 1/2.
21	A	부신수질.
22	C	염기 + 5탄당 + 인산.
23	D	IgG.
24	E	집합관에 아쿠아포린 삽입 → 물 투과성↑.
25	A	감수분열 산물 n=24 → 모세포 2n=48.
26	C	주사슬 C=O와 N–H 사이 수소결합.
27	A	β-글로빈 유전자 단일 염기 치환(Glu→Val).
28	A	저농도 용액 → 고농도 용액 쪽으로 물 이동.
29	B	접합(conjugation).
30	C	침샘 아밀레이스.
31	E	150×3 + 종결코돈 3 = 453.
32	D	적응방산.
33	A	2 mol × 58.5 = 117 g.
34	D	2(+1) + S + 3(−2) = 0 → +4.
35	C	남는 H^+ 0.1 mol/L·V, 부피 2V → 0.05 mol/L.

번호	정답	해설
36	B	주기에서 오른쪽으로 갈수록 증가.
37	C	C–H 4 + C–C σ 1 = 5σ, π 1.
38	C	2:3 → 6 mol Al : 9 mol Cl_2.
39	C	에스터화(피셔 에스터화).
40	C	0.05 mol / 0.25 L = 0.2 mol/L.
41	D	분자량 클수록 분산력↑.
42	A	$NH_3 + H^+ \rightarrow NH_4^+$.
43	D	$K \ll 1$ → 반응물 쪽 우세.
44	D	메틸아민. E는 아마이드.
45	C	최외각 7개 → 17족(Cl).
46	A	O(−1) → H_2O(−2) 환원, O_2(0) 산화 → 불균등화.
47	E	비편재화된 π 전자를 가진 방향족 고리.
48	B	200 m / 20 s = 10 m/s.
49	E	$\frac{1}{2}kx^2 = \frac{1}{2}\times200\times0.01 = 1$ J.
50	B	4∥12 = 3 Ω, +2 = 5 Ω.
51	A	$\lambda = v/f$.
52	E	100 × 0.5/0.01 = 5000 N.
53	E	$v^2/r = 16/2 = 8$.
54	D	0.06 kW × 5 h = 0.3 kWh.
55	E	$2^5 \rightarrow x+1 = 5$.
56	B	cos = 4/5 → tan = 3/4.
57	E	4! = 24.
58	D	기울기 3, 절편 −1.
59	A	$-2x > 4 \rightarrow x < -2$.
60	B	4번 반감 → 1/16.